Midterm Exam — Solutions

Lectures 1–2: Linear Algebra and Analytic Geometry

Problem 1

(a) Evaluation at 1 and differentiation are linear, so $T(p+q) = T(p) + T(q)$ and $T(cp) = cT(p)$. The columns are $T(1) = (1,0)$, $T(x) = (1,1)$, $T(x^2) = (1,2)$, $T(x^3) = (1,3)$:

$$[T]_{\mathcal{C},\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix}.$$

(b) The matrix has rank 2, so $\dim \operatorname{im}(T) = 2$ and $\operatorname{im}(T) = \mathbb{R}^2$. A polynomial with $p(1) = p'(1) = 0$ has a double root at 1, so $\ker(T) = \{(x-1)^2(c_0 + c_1 x)\}$, with the basis $(x-1)^2, (x-1)^3$ (coordinates $(1,-2,1,0)$ and $(-1,3,-3,1)$). Rank–nullity: $2 + 2 = 4 = \dim V$.

(c) By Theorem (quotient dimension), $\dim V/K = 4 - 2 = 2$, so it suffices to show independence: if $a(1+K) + b(x+K) = K$, then $a + bx \in K$, so $T(a+bx) = (a+b, b) = (0,0)$ and $a = b = 0$. For $x^3 + K = a(1+K) + b(x+K)$ we need $x^3 - a - bx \in K$, i.e., $T(x^3) = aT(1) + bT(x)$: $(1,3) = (a+b, b)$, so $b = 3$, $a = -2$, and

$$x^3 + K = -2\,(1+K) + 3\,(x+K). \qquad (\text{Check: } x^3 - 3x + 2 = (x-1)^2(x+2) \in K.)$$

Equivalently, $\bar{T} : V/K \to \mathbb{R}^2$ of the first isomorphism theorem maps $1 + K \mapsto (1,0)$ and $x + K \mapsto (1,1)$, a basis of $\mathbb{R}^2$.

(d) Both functionals vanish on K by the definition of K, so $\varphi_1, \varphi_2 \in K^\circ$. They are linearly independent: $\varphi_1(1) = 1$, $\varphi_2(1) = 0$, $\varphi_1(x-1) = 0$, $\varphi_2(x-1) = 1$. Since $K^\circ \cong (V/K)^*$ (quotients and duality), $\dim K^\circ = \dim V/K = 2$, so they form a basis.

Problem 2

(a) $M = aI_n + (b\mathbf{1})\mathbf{1}^\top$. By the determinant lemma, $\det(M) = \big(1 + \mathbf{1}^\top (aI)^{-1} b\mathbf{1}\big) \det(aI_n) = (1 + \frac{nb}{a}) a^n = a^{n-1}(a + nb)$.

(b) By Sherman–Morrison with $A = aI$, $\mathbf{u} = b\mathbf{1}$, $\mathbf{v} = \mathbf{1}$ (the denominator is $1 + \frac{nb}{a} \neq 0$):

$$M^{-1} = \frac{1}{a} I_n - \frac{\frac{b}{a^2}\mathbf{1}\mathbf{1}^\top}{1 + \frac{nb}{a}} = \frac{1}{a}\Big(I_n - \frac{b}{a+nb}\,\mathbf{1}\mathbf{1}^\top\Big).$$

(c) $M\mathbf{1} = (a+nb)\mathbf{1}$, and $M\mathbf{x} = a\mathbf{x}$ for every $\mathbf{x} \perp \mathbf{1}$, an $(n-1)$-dimensional subspace. Since $b \neq 0$, the eigenvalues $a + nb$ and a are distinct, with geometric multiplicities 1 and $n-1$; these add up to n, so the algebraic multiplicities are the same, M is diagonalizable (indeed symmetric), and $\det(M) = a^{n-1}(a+nb)$ is the product of the eigenvalues. The minimal polynomial is $m_M(t) = (t-a)(t-a-nb)$, a product of distinct linear factors.

(d) M is symmetric, so it is positive definite if and only if all its eigenvalues are positive: $a > 0$ and $a + nb > 0$. (For $b = 0$, this is $a > 0$, which is also correct.)

Problem 3

(a) Expanding $\det(A-\lambda I)$ and $\det(B-\lambda I)$ gives in both cases

$$p(\lambda) = -\lambda^3 + 4\lambda^2 - 5\lambda + 2 = -(\lambda-1)^2(\lambda-2).$$

(Check: $\operatorname{tr} = 4 = 1+1+2$ and $\det = 2 = 1\cdot 1\cdot 2$ for both.) The eigenvalues are 1 and 2.

(b) For both matrices, $\lambda = 1$ has algebraic multiplicity 2 and $\lambda = 2$ algebraic multiplicity 1 (hence geometric multiplicity 1). The geometric multiplicity of 1 is $3 - \operatorname{rank}(X - I)$:

$$A - I = \begin{pmatrix} 2 & -2 & -1 \\ 2 & -2 & -1 \\ -1 & 1 & 1 \end{pmatrix},\ \operatorname{rank} 2; \qquad B - I = \begin{pmatrix} 1 & -1 & -1 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix},\ \operatorname{rank} 1.$$

So $\lambda = 1$ has geometric multiplicity 1 for A (eigenvector $(1,1,0)^\top$) and 2 for B (eigenvectors $(1,1,0)^\top$, $(1,0,1)^\top$). B is diagonalizable, A is not.

(c) m divides p and has both eigenvalues as roots, so it is $(t-1)(t-2)$ or $(t-1)^2(t-2)$. Since B is diagonalizable, $m_B(t) = (t-1)(t-2)$; since A is not, $m_A(t) = (t-1)^2(t-2)$ (one can also check $(A-I)(A-2I) \neq 0$ and $(B-I)(B-2I) = 0$). Similar matrices have the same minimal polynomial (or: $B = PDP^{-1}$ would make A diagonalizable as well), so A and B are not similar, although they have the same characteristic polynomial, trace and determinant.

(d) Cayley–Hamilton: $A^3 - 4A^2 + 5A - 2I = 0$, so $A(A^2 - 4A + 5I) = 2I$ and

$$A^{-1} = \tfrac{1}{2}(A^2 - 4A + 5I) = \tfrac{1}{2}\begin{pmatrix} -1 & 3 & 1 \\ -3 & 5 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

From m_B: $B^2 - 3B + 2I = 0$, so $B(3I - B) = 2I$ and $B^{-1} = \frac{1}{2}(3I - B)$.

Problem 4

(a) Under $\mathbb{R}^2 \otimes \mathbb{R}^2 \cong \mathbb{R}^{2\times 2}$, $\mathbf{e}_i \otimes \mathbf{e}_j \mapsto \mathbf{e}_i\mathbf{e}_j^\top$, the tensor rank is the matrix rank. $t \mapsto \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has rank 1, and $t = (\mathbf{e}_1 + \mathbf{e}_2) \otimes (\mathbf{e}_1 + \mathbf{e}_2)$ by bilinearity. $s \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ has rank 2, so s is not a pure tensor.

(b) The map $\beta : V \times W \to W \otimes V$, $\beta(v, w) = w \otimes v$, is bilinear, since $\otimes$ is bilinear in each argument. By the universal property, there is exactly one linear $\sigma = \tilde{\beta}$ with $\sigma(v \otimes w) = w \otimes v$. In the same way, there is a linear $\tau : W \otimes V \to V \otimes W$ with $\tau(w \otimes v) = v \otimes w$. Then $\tau \circ \sigma$ is linear and fixes all pure tensors, which span $V \otimes W$, so $\tau \circ \sigma = \mathrm{id}$; similarly $\sigma \circ \tau = \mathrm{id}$. Hence σ is an isomorphism with inverse τ. (Uniqueness also follows directly: a linear map is determined by its values on the pure tensors, which span.)

(c) The change of basis matrix has the new basis vectors as columns, $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. The linear map (type $(1,1)$) becomes $P^{-1}AP = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ with trace 0; the

bilinear form (type $(0,2)$) becomes $P^\top AP = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ with trace 1. The trace of the linear map is basis independent: it is the contraction $C : V \otimes V^* \to \mathbb{R}$, defined without bases, and $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A)$. There is no contraction on $V^* \otimes V^*$, and indeed the trace of the bilinear form changed from 0 to 1.

(d) (i) $A_{ij}B_{ij}$, `np.einsum("ij,ij->", A, B)`. (ii) $q_b = X_{bi}M_{ij}X_{bj}$, `np.einsum("bi,ij,bj->b", X, M, X)`.

Problem 5

(a) $\|1\|^2 = 2$, so $e_1 = \frac{1}{\sqrt{2}}$. $\langle x, e_1\rangle = 0$ (odd integrand) and $\|x\|^2 = \frac{2}{3}$, so $e_2 = \sqrt{\frac{3}{2}}\,x$. For x^2: $\langle x^2, e_1\rangle e_1 = \frac{1}{2}\int_{-1}^{1} x^2\,dx = \frac{1}{3}$ and $\langle x^2, e_2\rangle = 0$, so $u_3 = x^2 - \frac{1}{3}$, with $\|u_3\|^2 = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}$, and $e_3 = \sqrt{\frac{45}{8}}(x^2 - \frac{1}{3})$.

(b) $p = \pi_U(f) = \sum_i \langle f, e_i\rangle e_i$. With the orthogonal (not normalized) basis $1, x, x^2 - \frac{1}{3}$: $\frac{\langle |x|,1\rangle}{\langle 1,1\rangle} = \frac{1}{2}$; $\langle |x|, x\rangle = 0$ (odd); $\langle |x|, x^2 - \frac{1}{3}\rangle = 2\int_0^1 (x^3 - \frac{x}{3})dx = \frac{1}{6}$, divided by $\frac{8}{45}$ gives $\frac{15}{16}$. So

$$p(x) = \frac{1}{2} + \frac{15}{16}\Big(x^2 - \frac{1}{3}\Big) = \frac{15x^2 + 3}{16}.$$

(c) By Pythagoras, $\|f - p\|^2 = \|f\|^2 - \|p\|^2 = \frac{2}{3} - \Big(\frac{1}{4}\cdot 2 + \frac{225}{256}\cdot\frac{8}{45}\Big) = \frac{2}{3} - \frac{21}{32} = \frac{1}{96}$.

(d) Let $\mathbf{x} \in \mathbb{R}^n$. Clearly $P\mathbf{x} \in \operatorname{col}(P)$. Every $\mathbf{u} \in \operatorname{col}(P)$ is $\mathbf{u} = P\mathbf{z}$, and

$$(\mathbf{x} - P\mathbf{x})^\top P\mathbf{z} = \mathbf{x}^\top P\mathbf{z} - \mathbf{x}^\top P^\top P\mathbf{z} = \mathbf{x}^\top P\mathbf{z} - \mathbf{x}^\top P^2\mathbf{z} = 0.$$

So $P\mathbf{x} \in \operatorname{col}(P)$ and $\mathbf{x} - P\mathbf{x} \in \operatorname{col}(P)^\perp$, which characterizes $\pi_{\operatorname{col}(P)}(\mathbf{x})$ uniquely.

Problem 6

(a) $A^\top A = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$ has the eigenvalues 45 and 5 with eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. So $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$, and $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$: $\mathbf{u}_1 = \frac{1}{\sqrt{10}}(1,3)^\top$, $\mathbf{u}_2 = \frac{1}{\sqrt{10}}(3,-1)^\top$. Hence

$$A = \underbrace{\frac{1}{\sqrt{10}}\begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{pmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{V^\top}.$$

(Signs of matching pairs $\mathbf{u}_i, \mathbf{v}_i$ may be flipped together.)

(b) $\|A\|_2 = \sigma_1 = 3\sqrt{5}$, $\|A\|_F = \sqrt{45 + 5} = \sqrt{50} = 5\sqrt{2}$ (check: $9+16+25 = 50$), $\kappa(A) = \sigma_1/\sigma_2 = 3$.

(c) By Eckart–Young, $A_1 = \sigma_1\mathbf{u}_1\mathbf{v}_1^\top = \frac{3}{2}\begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$, and $\|A - A_1\|_2 = \sigma_2 = \sqrt{5}$, $\|A - A_1\|_F = \sigma_2 = \sqrt{5}$ (only one singular value is left).

(d) If $C\mathbf{x} = \mathbf{0}$, then $C^\top C\mathbf{x} = \mathbf{0}$. Conversely, if $C^\top C\mathbf{x} = \mathbf{0}$, then $0 = \mathbf{x}^\top C^\top C\mathbf{x} = \|C\mathbf{x}\|^2$, so $C\mathbf{x} = \mathbf{0}$. Both matrices have n columns, so by rank–nullity $\operatorname{rank}(C^\top C) = n - \dim \operatorname{null}(C^\top C) = n - \dim \operatorname{null}(C) = \operatorname{rank}(C)$. The $n \times n$ matrix $C^\top C$ is invertible iff its rank is n, iff $\operatorname{rank}(C) = n$, iff the columns of C are linearly independent.